* Case - 2 :- $Q(x) = \sin(ax+b)$ or $\cos(ax+b)$.

i) If $Q(x) = \sin(ax+b)$ then eq^{n} (2) reduces to $f(D)y = \sin(ax+b)$.

Now, $D[\sin(ax+b)] = a\cdot\cos(ax+b)$

$D^2[\sin(ax+b)] = (-a^2)\sin(ax+b)$

$D^3[\sin(ax+b)] = -a^3\cdot\sin(ax+b)$

$D^4[\sin(ax+b)] = a^4\sin(ax+b)$

$(D^2)^2[\sin(ax+b)] = (-a^2)^2\sin(ax+b)$

In general, $(D^2)^n[\sin(ax+b)] = (-a^2)^n\sin(ax+b)$.

This shows that $\phi(D^2)\sin(ax+b) = \phi(-a^2)\sin(ax+b)$.

- Operating both the sides with $\frac{1}{\phi(D^2)}$,

$$\frac{1}{\phi(D^2)}\left[\phi(D^2)\sin(ax+b)\right] = \frac{1}{\phi(D^2)}\left[\phi(-a^2)\sin(ax+b)\right]$$

$$\therefore \sin(ax+b) = \phi(-a^2)\frac{1}{\phi(D^2)}\sin(ax+b).$$

$$\therefore \frac{1}{\phi(-a^2)}\sin(ax+b) = \frac{1}{\phi(D^2)}\sin(ax+b)$$

$$\therefore \frac{1}{\phi(D^2)}\sin(ax+b) = \frac{1}{\phi(-a^2)}\sin(ax+b)$$

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If $f(D) = \phi(D^2)$ then

$$P.I = \frac{1}{f(D)} \sin(ax+b)$$

$$= \frac{1}{\phi(D^2)} \sin(ax+b)$$

$$= \frac{1}{\phi(-a^2)} \sin(ax+b), \text{ if } \phi(-a^2) \neq 0.$$

- If $\phi(-a^2) = 0$ then D^2+a^2 is a factor of $\phi(D^2)$ and hence the above rule fails.

$$P.I = \frac{1}{\phi(D^2)} \sin(ax+b)$$

$$= \frac{1}{\phi(D^2)} [\text{I.P. of } e^{i(ax+b)}] \quad \{\text{Imaginary part}\}$$

$$= \text{I.P. of } \frac{1}{\phi(D^2)} e^{i(ax+b)}$$

$$= \text{I.P. of } x \cdot \frac{1}{\phi'(D^2)} e^{i(ax+b)}$$

$$= \text{I.P. of } x \cdot \frac{1}{\phi'(i^2a^2)} e^{i(ax+b)}$$

$$= \text{I.P. of } x \cdot \frac{1}{\phi'(-a^2)} e^{i(ax+b)}$$

$$= x \cdot \frac{1}{\phi'(-a^2)} \sin(ax+b)$$

- If $\phi'(-a^2) = 0$ then $\frac{1}{\phi(D^2)} \sin(ax+b) = x^2 \cdot \frac{1}{\phi''(-a^2)} \sin(ax+b)$, where $\phi''(-a^2) \neq 0$.

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In general, if $\phi^{n}(-a^2) = 0$ then

$$P.I. = \frac{1}{\phi(D^2)} \sin(ax+b)$$

$$= x^{n+1} \frac{1}{\phi^{(n+1)}(-a^2)} \sin(ax+b), \text{ where } \phi^{(n+1)}(-a^2) \neq 0.$$

ii) Similarly, if $Q(x) = \cos(ax+b)$

$$P.I. = \frac{1}{\phi(D^2)} \cos(ax+b)$$

$$= \frac{1}{\phi(-a^2)} \cos(ax+b), \quad \phi(-a^2) \neq 0.$$

- If $\phi(-a^2) = 0$ then $P.I. = \frac{1}{\phi(D^2)} \cos(ax+b)$

$$= x \cdot \frac{1}{\phi'(-a^2)} \cos(ax+b)$$

- If $\phi'(-a^2) = 0$ then

$$P.I = \frac{1}{\phi(D^2)} \cos(ax+b)$$

$$= x^2 \cdot \frac{1}{\phi''(-a^2)} \cos(ax+b),$$

where $\phi''(-a^2) \neq 0$.

- In general, if $\phi^{(n)}(-a^2) = 0$ then

$$P.I. = \frac{1}{\phi(D^2)} \cos(ax+b)$$

$$= x^{n+1} \frac{1}{\phi^{(n+1)}(-a^2)} \cos(ax+b), \text{ where } \phi^{(n+1)}(-a^2) \neq 0.$$

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* Note :-

If after replacing D^2 by $-a^2$, $f(D)$ contains terms of D then the denominator is rationalized to obtain the even powers of D.

* Examples :-

(1) Solve $(D^2+9)y = \sin 4x$

→

The A.E. is $m^2+9=0$

$\therefore m = \pm 3i$ (complex)

$\therefore C.F. = e^{0x}(C_1 \cos 3x + C_2 \sin 3x)$

$P.I. = \frac{1}{D^2+9} \sin 4x$

$= \frac{1}{-4^2+9} \sin 4x$

$= -\frac{1}{7} \sin 4x$

Hence, the G.S. is $y = C.F. + P.I.$

$\therefore y = C_1 \cos 3x + C_2 \sin 3x - \frac{1}{7} \sin 4x.$

(2) Solve $(D^2+1)y = \sin^2 x.$

→

A.E. is $m^2+1=0$

$\therefore m = \pm i$

$C.F. = e^{0x}(C_1 \cos x + C_2 \sin x)$

$P.I. = \frac{1}{D^2+1} \sin^2 x$

$= \frac{1}{D^2+1}\left(\frac{1-\cos 2x}{2}\right)$

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$= \frac{1}{D^2+1} \frac{1}{2} = \frac{1}{2} \frac{1}{D^2+1} \cos 2x$

$= \frac{1}{2} \frac{1}{D^2+1} e^{0x} = \frac{1}{2} \frac{1}{-2^2+1} \cos 2x$

$= \frac{1}{2} \frac{1}{0+1} e^{0x} = \frac{1}{2} \left(\frac{-1}{3}\right) \cos 2x$

$= \frac{1}{2} + \frac{1}{6} \cos 2x.$

(3) Solve $(D^2+3D+2)y = \sin 2x.$

→

The A.E. is $m^2+3m+2=0$

$(m+1)(m+2)=0$

$\therefore m = -1, -2.$

$C.F. = C_1 e^{-x} + C_2 e^{-2x}$

$P.I. = \frac{1}{D^2+3D+2} \sin 2x$

$= \frac{1}{-2^2+3D+2} \sin 2x$

$= \frac{1}{3D-2} \sin 2x$

$= \frac{1}{(3D-2)} \frac{(3D+2)}{(3D+2)} \sin 2x$

$= \frac{3D+2}{9D^2-4} \sin 2x$

$= \frac{3D+2}{9(-2^2)-4} \sin 2x$

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$$P.I = \frac{3D+2}{-40} \sin 2x$$

$$= \frac{-1}{40}\left[3(D\sin 2x) + 2(\sin 2x)\right]$$

$$= \frac{-1}{40}\left[6\cos 2x + 2\sin 2x\right]$$

$$= \frac{-1}{20}(3\cos 2x + \sin 2x).$$

- The G.S. is $y = Ce^{-x} + Ce^{-2x} - \frac{1}{20}(3\cos 2x + \sin 2x)$.

④ Solve $(D^2+9)y = 2\sin 3x + \cos 3x$

→

A.E. is $m^2+9=0$

$\therefore m = \pm 3i$

$\therefore C.F. = C_1 \cos 3x + C_2 \sin 3x.$

$$P.I. = \frac{1}{D^2+9}(2\sin 3x + \cos 3x)$$

$$= 2\,\frac{1}{D^2+9}\sin 3x + \frac{1}{D^2+9}\cos 3x$$

$$= 2 \cdot x \cdot \frac{1}{2D}\sin 3x + \frac{x}{2D}\cos 3x$$

$$= x\int \sin 3x\,dx + \frac{x}{2}\int \cos 3x\,dx$$

$$= x\left(\frac{-\cos 3x}{3}\right) + \frac{x}{2}\left(\frac{\sin 3x}{3}\right)$$

$$= -\frac{x}{3}\cos 3x + \frac{x}{6}\sin 3x.$$

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- The G.S. is

$$y = C_1 \cos 3x + C_2 \sin 3x - \frac{x}{3}\cos 3x + \frac{x}{6}\sin 3x.$$

(5) Solve $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} + 3y = \sin 3x \cdot \cos 2x$

$\Rightarrow$

The A.E. is $m^2 - 4m + 3 = 0$

$(m-1)(m-3) = 0$

$m = 1, 3$

$\therefore$ C.F. $= C_1 e^{x} + C_2 e^{3x}$.

$$P.I. = \frac{1}{D^2-4D+3}(\sin 3x \cdot \cos 2x)$$

$$= \frac{1}{D^2-4D+3}\left(\frac{\sin 5x + \sin x}{2}\right)$$

$$= \frac{1}{2}\frac{1}{D^2-4D+3}\sin 5x + \frac{1}{2}\frac{1}{D^2-4D+3}\sin x.$$

$$= \frac{1}{2}\frac{1}{-5^2-4D+3}\sin 5x + \frac{1}{2}\frac{1}{-1^2-4D+3}\sin x$$

$$= \frac{1}{2}\frac{1}{-4D-22}\sin 5x + \frac{1}{2}\frac{1}{2-4D}\sin x$$

$$= \frac{-1}{4}\frac{1}{2D+11}\frac{2D-11}{2D-11}\sin 5x + \frac{1}{4}\frac{1}{2-4D}\frac{1+2D}{1+2D}\sin x$$

$$= \frac{-1}{4}\frac{2D-11}{4D^2-121}\sin 5x + \frac{1}{4}\frac{1+2D}{1-4D^2}\sin x.$$

$$= \frac{-1}{4}\frac{2D-11}{4(-5^2)-121}\sin 5x + \frac{1}{4}\frac{1+2D}{1-4(-1^2)}\sin x.$$

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$$P.I. = \frac{-1}{-221}\frac{2D-11}{}\sin 5x + \frac{1}{4}\frac{1+2D}{5}\sin x.$$

$$= \frac{1}{884}(2D-11)\sin 5x + \frac{1}{20}(1+2D)\sin x.$$

$$= \frac{2}{884}(D\sin 5x) - \frac{11}{884}\sin 5x + \frac{1}{20}\sin x + \frac{2}{20}(D\sin x)$$

$$= \frac{5}{442}\cos 5x - \frac{11}{884}\sin 5x + \frac{1}{20}\sin x + \frac{1}{10}\cos x.$$

∴ The G.S. is $y = C_1 e^{x} + C_2 e^{3x} + \frac{5}{442}\cos 5x - \frac{11}{884}\sin 5x + \frac{1}{20}\sin x + \frac{1}{10}\cos x.$

(6) Solve $(D-1)^2(D^2+1)y = e^x + \sin^2\frac{x}{2}$.

→

A.E. is $(m-1)^2(m^2+1) = 0$

$\therefore (m-1)^2 = 0, \quad m^2+1 = 0$

$\therefore m = 1, 1, \pm i$

$\therefore C.F. = e^{x}(C_1 + C_2 x) + e^{0x}(C_3\cos x + C_4\sin x)$

$$P.I. = \frac{1}{(D-1)^2(D^2+1)}\cdot e^x + \sin^2\frac{x}{2}$$

$$= \frac{1}{(D-1)^2(D^2+1)}\left(e^x + \frac{1-\cos x}{2}\right)$$

$$= \frac{1}{(D-1)^2(D^2+1)}\left(e^x + \frac{1}{2}e^{0x} - \frac{1}{2}\cos x\right)$$

$$= \frac{1}{(D-1)^2(D^2+1)}\cdot e^x + \frac{1}{2(D-1)^2(D^2+1)}e^{0x} - \frac{1}{2(D+1)^2(D^2+1)}\cos x.$$

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$$P.I. = \frac{1}{(D-1)^2}\frac{1}{(i^2+1)}e^x + \frac{1}{2(D-1)^2(D+1)}e^{0x} - \frac{1}{(D^2+1)}\frac{1}{(D^2-2D+1)}\frac{\cos x}{2}$$

$$= \frac{1}{2}\frac{1}{(D-1)^2}e^x + \frac{1}{2} - \frac{1}{2(D^2+1)[(-1^2)-2D+1)]}\cos x.$$

$$= \frac{x}{2}\frac{1}{2(D-1)}e^x + \frac{1}{2} - \frac{1}{2}\frac{1}{(D^2+1)(-2D)}\cos x$$

$$= \frac{x^2}{4}e^x + \frac{1}{2} + \frac{1}{4}\frac{1}{D^2+1}\frac{1}{D}\cos x.$$

$$= \frac{x^2e^x}{4} + \frac{1}{2} + \frac{1}{4}\frac{1}{D^2+1}\sin x$$

$$= \frac{x^2e^x}{4} + \frac{1}{2} + \frac{1}{4}x\frac{1}{2D}\sin x$$

$$= \frac{x^2e^x}{4} + \frac{1}{2} + \frac{1}{8}x(-\cos x)$$

$$= \frac{x^2e^x}{4} + \frac{1}{2} - \frac{x}{8}\cos x$$

∴ The G.S. is $y = e^x(C_1 + C_2x) + C_3\cos x + C_4\sin x + \frac{x^2e^x}{4} + \frac{1}{2} - \frac{x}{8}\cos x.$

* Case - 3 $Q(x) = x^m$.

Here, $f(D)y = x^m$

$$\therefore P.I. = \frac{1}{f(D)}x^m$$

$$= [f(D)]^{-1}x^m$$

$$= [1+\phi(D)]^{-1}x^m.$$

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Expanding in ascending powers of D upto D^m using Binomial Expansion, since $D^n x^m = 0$, where $n > m$.

$$P.I. = (a_0 + a_1 D + a_2 D^2 + \cdots + a_m D^m)\, x^m$$

* **Examples:-**

(1) Solve $(D^2 + 2D + 1)\, y = x$.

→

The A.E. is $m^2 + 2m + 1 = 0$

$(m+1)^2 = 0$

$\therefore m = -1, -1$

$$C.F. = e^{-x}(C_1 + C_2 x)$$

$$P.I. = \frac{1}{D^2 + 2D + 1}\, x$$

$$= \frac{1}{(1+D)^2}\, x$$

Expansion of $(1+x)^m = 1 + mx + \frac{m(m-1)}{2!}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \cdots$

$$= (1+D)^{-2}\, x \qquad m = -m$$

$$(1+x)^{-m} = 1 - mx + \frac{(-m)(-m-1)}{2!}x^2 + \frac{(-m)(-m-1)(-m-2)}{3!}x^3 + \cdots$$

$$= 1 - 2D + \frac{2(2+1)}{2!}D^2 - \cdots)x$$

$$= (1 - 2D + 3D^2 - \cdots)x$$

$$= x - 2Dx + 3D^2 x - \cdots \qquad = 1 - mx + \frac{m(m+1)}{2!}x^2 - \frac{m(m+1)(m+2)}{3!}x^3 + \cdots$$

$$= x - 2 + 0$$

$$= x - 2.$$

The G.S. is $y = e^{-x}(C_1 + C_2 x) + x - 2$.

(2) Solve $y'' + 2y' + 3y = 2x^2$.

→

The A.E. is $m^2 + 2m + 3 = 0$.

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$$m = \frac{-2 \pm \sqrt{4-12}}{2} = \frac{-2 \pm \sqrt{-8}}{2} = \frac{-2 \pm 2\sqrt{2}i}{2} = -1 \pm \sqrt{2}i$$

$$C.F. = e^{-x}(C_1 \cos\sqrt{2}x + C_2 \sin\sqrt{2}x).$$

$$P.I. = \frac{1}{D^3+2D+3}(2x^2)$$

$$= \frac{1}{3\left[1 + \frac{D^3+2D}{3}\right]}(2x^2)$$

$$= \frac{2}{3}\left[1 + \frac{D^3+2D}{3}\right]^{-1} x^2$$

$$= \frac{2}{3}\left[1 - \left(\frac{D^3+2D}{3}\right) + \left(\frac{D^3+2D}{3}\right)^2 - \cdots\right] x^2$$

$$[(1+x)^{-1} = 1 - x + x^2 - \cdots]$$

$$= \frac{2}{3}\left[x^2 - \frac{1}{3}D^2x^2 - \frac{2}{3}Dx^2 + \frac{4}{9}D^2x^2 - \cdots\right]$$

$$= \frac{2}{3}\left[x^2 - \frac{2}{3} - \frac{2}{3}(2x) + \frac{4}{9}\cdot 2 - \cdots\right]$$

$$= \frac{2}{3}\left(x^2 - \frac{4}{3}x + \frac{2}{9}\right).$$

Hence, the G.S. is $y = e^{-x}(C_1 \cos\sqrt{2}x + C_2 \sin\sqrt{2}x) + \frac{2}{3}\left(x^2 - \frac{4}{3}x + \frac{2}{9}\right)$.

(3) Solve $(D^2+D)y = x^2+2x+4$

→

The A.E. is $m^2+m = 0$

$m(m+1) = 0$

$\therefore m = 0, -1.$

$\therefore C.F. = C_1e^{0x} + C_2e^{-x}.$

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$$P.I. = \frac{1}{D^2+D}(x^2+2x+4)$$

$$= \frac{1}{D(D+1)}(x^2+2x+4)$$

$$= \frac{1}{D}(1+D)^{-1}(x^2+2x+4)$$

$$= \frac{1}{D}(1-D+D^2-D^3+\dots)(x^2+2x+4)$$

$$= \frac{1}{D}\left[x^2+2x+4 - D(x^2+2x+4) + D^2(x^2+2x+4) - D^3(x^2+2x+4) + \dots\right]$$

$$= \frac{1}{D}\left[x^2+2x+4-(2x+2)+2-0\right]$$

$$= \frac{1}{D}(x^2+4)$$

$$= \int (x^2+4)\,dx$$

$$= \frac{x^3}{3} + 4x$$

Hence, the G.S. is $y = C_1 + C_2 e^{-x} + \frac{x^3}{3} + 4x$.

(4) Solve $(D^2+2)y = x^3+x^2+e^{-2x}+\cos 3x$.

→

A.E. is $m^2+2=0$

$\therefore m^2 = -2$

$\therefore m = \pm i\sqrt{2}$

$\therefore C.F. = e^{0x}(C_1\cos\sqrt{2}x + C_2\sin\sqrt{2}x)$

$$P.I. = \frac{1}{D^2+2}(x^3+x^2+e^{-2x}+\cos 3x)$$

$$= \frac{1}{2\left(1+\frac{D^2}{2}\right)}(x^3+x^2) + \frac{1}{D^2+2}e^{-2x} + \frac{1}{D^2+2}\cos 3x$$

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$$P.I. = \frac{1}{2}\left(1+\frac{D^2}{2}\right)^{-1}(x^3+x^2) + \frac{1}{(-2)^2+2}e^{-2x} + \frac{1}{-3^2+2}\cos 3x.$$

$$= \frac{1}{2}\left(1-\frac{D^2}{2}+\frac{D^4}{4}-\cdots\right)(x^3+x^2) + \frac{1}{6}e^{-2x} + \frac{1}{(-7)}\cos 3x.$$

$$= \frac{1}{2}\left[(x^3+x^2) - \frac{1}{2}D^2(x^3+x^2) + \frac{1}{4}D^4(x^3+x^2) - \cdots\right] + \frac{1}{6}e^{-2x}$$

$$-\frac{1}{7}\cos 3x$$

$$= \frac{1}{2}\left[x^3+x^2 - \frac{1}{2}(6x+2) + 0\right] + \frac{1}{6}e^{-2x} - \frac{1}{7}\cos 3x$$

$$= \frac{1}{2}(x^3+x^2-3x-1) + \frac{1}{6}e^{-2x} - \frac{\cos 3x}{7}$$

Hence, the G.S. is

$$y = C_1\cos\sqrt{2}x + C_2\sin\sqrt{2}x + \frac{1}{2}(x^3+x^2-3x-1) + \frac{e^{-2x}}{6} - \frac{\cos 3x}{7}$$

(5) Solve $(D^3-D^2-6D)y = 1+x^2$.

→

The A.E. is $m^3-m^2-6m=0$

$\therefore m(m^2-m-6)=0$

$\therefore m(m-3)(m+2)=0$

$\therefore m = 0, 3, -2.$

$$C.F. = C_1e^{0x} + C_2e^{3x} + C_3e^{-2x}.$$

$$P.I. = \frac{1}{D^3-D^2-6D}(1+x^2).$$

$$= \frac{1}{-6D\left[1-\frac{D^2-D}{6}\right]}(1+x^2)$$

$$= \frac{-1}{6D}\left[1-\left(\frac{D^2-D}{6}\right)\right]^{-1}(1+x^2)$$

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$$I.=\frac{-1}{6D}\left[1+\left(\frac{D^2-D}{6}\right)+\left(\frac{D^2-D}{6}\right)^2+\cdots\right](1+x^2)$$

$$=\frac{-1}{6D}\left[1+\frac{D^2-D}{6}+\frac{D^4-2D^3+D^2}{36}+\cdots\right](1+x^2)$$

$$=\frac{-1}{6D}\left[1-\frac{D}{6}+\frac{7D^2}{36}-\frac{D^3}{18}+\cdots\right](1+x^2)$$

$$=\frac{-1}{6D}\left[(1+x^2)-\frac{1}{6}D(1+x^2)+\frac{7}{36}D^2(1+x^2)-\frac{1}{18}D^3(1+x^2)+\cdots\right]$$

$$=\frac{-1}{6D}\left[1+x^2-\frac{1}{6}(2x)+\frac{7}{36}(2)-0\right]$$

$$=\frac{-1}{6D}\left[1+x^2-\frac{1}{3}x+\frac{7}{18}\right]$$

$$=\frac{-1}{6D}\left[x^2-\frac{1}{3}x+\frac{25}{18}\right]$$

$$=\frac{-1}{6}\left[\frac{x^3}{3}-\frac{x^2}{6}+\frac{25}{18}x\right]=\frac{-1}{18}\left(x^3-\frac{x^2}{2}+\frac{25}{6}x\right)$$

The G.S. is $y=C_1+C_2e^{3x}+C_3e^{-2x}-\frac{1}{18}\left(x^3-\frac{x^2}{2}+\frac{25}{6}x\right)$.

(6) Solve $(D^4-16)y=e^{2x}+x^4$, where $D=\frac{d}{dx}$.

→

The A.E. is $m^4-16=0$

$\therefore (m^2-4)(m^2+4)=0$

$\therefore (m+2)(m-2)=0,\ m^2+4=0$

$m=-2,2,\ m^2=-4$

$m=-2,2,\ \pm 2i$

C.F. $=C_1e^{-2x}+C_2e^{2x}+e^{0x}(C_3\cos 2x+C_4\sin 2x)$.

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$$P.I. = \frac{1}{D^4-16}(e^{2x} + x^4)$$

$$= \frac{1}{D^4-16} e^{2x} + \frac{1}{D^4-16} x^4$$

$$= x\frac{1}{4D^3} e^{2x} + \frac{1}{-16\left[1 - \frac{D^4}{16}\right]} x^4$$

$$= \frac{x}{4}\frac{1}{2^3} e^{2x} - \frac{1}{16}\left[1 - \frac{D^4}{16}\right]^{-1} x^4$$

$$= \frac{x}{32} e^{2x} - \frac{1}{16}\left[1 + \frac{D^4}{16} + \cdots\right] x^4$$

$$= \frac{x}{32} e^{2x} - \frac{1}{16}\left[x^4 + \frac{1}{16} D^4 x^4\right].$$

$$= \frac{x}{32} e^{2x} - \frac{1}{16}\left[x^4 + \frac{1}{16} 24\right]$$

$$= \frac{x}{32} e^{2x} - \frac{x^4}{16} - \frac{3}{32}$$

$$= \frac{1}{32}(xe^{2x} - 2x^4 - 3)$$

Hence, the G.S. is $y = C_1 e^{2x} + C_2 e^{-2x} + C_3 \cos 2x + C_4 \sin 2x + \frac{1}{32}(xe^{2x} - 2x^4 - 3)$.

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